

Portfolio Project:

Hullabaloo Party Supplies Customer Order System

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Executive Summary

Hullabaloo Party Supplies (HPS), a party supply provider and distributor with thirteen nationwide locations, is shifting from a Selling to Vendors, a B2B business model, to a Selling directly to Consumers (B2C) business model. HPS management has stated that the existing B2B Information System (B2B-IS) does not support their new B2C business model as it lacks the proper customer interface and functionality. This proposal recommends implementing a new B2C-IS, more specifically a Customer Order System (COS) that is more suited for HPS's new business goals. The proposal is part of the Project Initiation and Planning (PIP) process for the development of a B2C-IS at HPS. In the early phase of the PIP, HPS management stated that the COS needs to allow customers to order multiple different items online in chosen quantities, provide complete customer information, create an account, see calculated order costs including tax, pay via credit card or PayPal, have ordered items debited from inventory, and choose from multiple shipping options. (Note: Once an order is complete, the process will transition to the B2B-IS shipping system.) This initial proposal outlines an overview of the purpose of the project, the problem statement, goals and objectives, key assumptions, project stakeholders, a high-level Work Breakdown Structure (WBS) for requirements gathering, and potential risks for the new COS project. It also provides a detailed list of requirements, a high-level Data Flow Diagram (DFD) of the system, a feasibility economic analysis, and proposes to utilize the Agile Scrum methodology to develop the system. Additionally, this proposal provides implementation, testing, and training plans, which are crucial for the successful deployment, user adoption, and effective functionality of the new COS.

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Goals and Objectives

HPS is a party supply provider and distributor selling to third-party vendors with thirteen nationwide locations, and has recently decided to shift from a B2B business model to a B2C business model. This will allow consumers to order from HPS directly, and HPS will then ship directly to them, eliminating the need for third-party vendors. After reviewing the existing B2B-IS, management has identified that the system does not meet the requirements of the B2C business model. Therefore, a new system is needed in the form of a B2C-IS that specifically supports the B2C business model. Two systems have been identified as being part of the B2C-IS: the Customer Order System (COS) and the Party Supply Shipping System. This proposal only addresses the COS part of the B2C-IS. The purpose of this proposal is to initiate the planning and development of COS suited to meet the needs of HPS's new B2C goals. The proposal is part of the PIP phase, which is a critical activity in the development project (Valacich & George, 2020a). It addresses the initial assessment of the project's scope, feasibility, potential risks, and resource requirements necessary for planning the COS development. Furthermore, to plan and assess the feasibility of the COS is essential to define its expected functionality.

The COS needs to function as a core B2C-IS component, meeting the following requirements provided by HPS management:

- Gather customer information.
- Manage customer accounts.
- Handle orders (including orders composed of multiple items).
- Manage payment processing.
- Manage inventory levels based on orders.
- Present customer shipping choices (once an order is complete, the process will transition to the shipping system).

This functionality will enable HPS to directly sell party supplies to consumers via an e-commerce store system (the new COS), bypassing third-party vendors. Moreover, aligning COS development with HPS business objectives, particularly the shift from B2B-IS to B2C-IS, is a critical factor for accomplishing HPS's mission successfully (Valacich & George, 2020b).

Problem Statement

It is important to understand how the existing B2B-IS conflicts with its HPS' business plans.

Understanding this conflict is essential for justifying the need for the proposed solution (Jaggi, 2023). To that purpose, this problem statement provides a clear description of the problem. HPS's existing B2B-IS is optimized for large-volume transactions with vendors and lacks the architecture, user interface, and functionalities required for a B2C e-commerce small-volume consumer transactions. In other words, the system is not optimized for personal customer accounts, processing consumer payment methods (PayPal), managing inventory based on B2C model needs, managing small orders typically associated with the B2C model, calculating sales tax for consumers, or providing a friendly User Interface (UI) and support for individual consumers, rendering the B2B-IS infeasible to meet the requirements associated with the new HPS business goals. Additionally, to meet these requirements, a new dedicated B2C-IS is needed, specifically, a new COS that is optimized for consumer online ordering operations. Without such a system, HPS will be unable to meet its new B2C business goals.

Key Assumptions

During the early PIP phase for the COS, the following key assumptions were made:

- "The project team members have the required skills and expertise to perform their assigned roles effectively" (Samartsheet, 2023).

- HPS management is committed to shifting from a B2B system to a B2C system and supports this project.
- “The project requirements and scope are well defined and will not significantly change during the project” (Samartsheet, 2023).
- Stakeholders (Marketing, IT, Finance, Warehouse) will support the project by being available for interviews, and they have a clear understanding of the project goals and objectives
- Funding and resources are planned to be allocated for the project.
- The existing inventory management system is accessible through an API or by other means.
- The shipping system of the new B2C-IS will integrate with the COS by receiving order data from the COS.

Project Stakeholders

The stakeholders for the HPS Customer Order System (COS) include:

- The HPS management department, they are the project sponsor, providing the funds, the COS requirements, and final approval.
- The sales and purchase department is responsible for setting item prices and defining promotional prices
- The marketing department defines customer experience requirements for the COS, the item presentation designs and running promotions.
- The IT department and the systems analyst are responsible for project management, analysis, design, development, implementation, and maintenance of the COS.
- The finance department is responsible for defining COS payment processing methods and tax calculation (equations).
- The warehouse (inventory Staff) defines how COS orders impact inventory, item availability, and how it should interface with the shipping system.
- Customers – the consumers are the end-users of the COS; their needs define the functionality requirements of the system.
- Legal and compliance teams are responsible for ensuring that the COS adheres to data privacy regulations and payment standards.

High-Level WBS for Requirements Gathering

WBS for an e-commerce project is a visual, hierarchical deconstruction of the project (Yakovlieva, 2024). It divides the project into manageable tasks (Valacich & George, 2020c). Requirements gathering ensures that the final version of COS meets stakeholder needs and avoids costly scope issues (Satpathy, 2024). Below is a six-step WBS for the COS requirements gathering:

1. Project Initiation & Planning Refinement
 - Confirm primary, secondary, and tertiary stakeholders identified in the early PIP phase. One or more analysts are assigned to work with the stakeholders (Valacich & George, 2020a)
 - Define engagement strategy for the analysis phase. This includes workshops and promoting communication between analysts and stakeholders.
 - Schedule initial analysis phase kick-off/review meetings by developing a preliminary schedule.
2. Requirements Elicitation
 - Stakeholder interviews need to be performed. This is done to understand the needs and expectations of the stakeholders, as well as to collect opinions and facts.
 - Facilitate requirements workshops (brainstorming, groups for UI). By conducting Joint Application Design (JAD) sessions with users, managers, and analysts to collect requirements and resolve potential conflicts (Valacich & George, 2020d).

- Perform document analysis (review B2C-IS documentation like business forms, reports, and manuals).
- Analyze system interfaces such as UI and payment gateways by observing and creating prototypes.
- 3. Requirements Documentation
 - Define documentation format utilizing user cases and stories.
 - Document functional requirements (ordering, shopping cart, account mgmt., payment), defining what the system needs to do.
 - Document non-functional requirements, describing how the system needs to function.
 - Document data requirements; for example, size and how the data needs to be handled.
 - Document interface requirements, how COS needs to interact with other systems.
- 4. Analysis of Requirements and Prioritization
 - Analyze documented requirements.
 - Model requirements by using Data Flow Diagrams (DFD).
 - Prioritize requirements based on value or urgency.
- 5. Validate Requirements
 - Make sure that the gathered requirements meet the needs of the stakeholders and are testable.
 - Perform review meetings/walk-throughs of the requirements (Valacich & George, 2020a).
 - Obtain stakeholder agreement/sign-off on baseline requirements.
- 6. Requirements Management Planning
 - Define a requirement change/modification control process to use during the development process.
 - Establish a requirements traceability (linking requirements to design, code, tests).
 - Configure requirements management tool/repository (e.g., GitHub)

COS Risks

Identifying risk is an essential activity of the PIP (Valacich & George, 2020a). The following is a list of risks for the COS project and their descriptions:

Economic Risks:

- Cost overruns in COS development and integration. In other words, the development and integration processes could overrun the budget due to unforeseen and/or unplanned expenses.
- Benefit shortfall as B2C sales volume may prove lower than projected, the benefits from COS may not be met.
- Total Cost of Ownership (TCO) was not calculated properly or failed to account for all the system's operational and maintenance expenses.

Technical Risks:

- Difficulty integrating the COS with the future shipping system or inventory system.
- COS performance degrades as customer volume grows, failure to plan growth accurately.
- Failure to implement adequate, secure customer personnel data.

Operational Risks:

- Consumers find the COS difficult to use, such as difficulty navigating or understanding the system.
- HPS staff experience significant difficulty adapting to the new B2C-IS COS. Inadequate training or not enough training.
- Issues maintaining inventory data may arise between the COS and warehouse systems.
- Issues with system downtime during the initial launch of the system.

Legal Risks:

- Failure of the COS to meet legal requirements for securing customer private information.

- Payment security is non-compliance with Payment Card Industry Data Security Standard.

Schedule Risks:

- Unrealistic timelines due to failure to identify potential delays in development, testing, or implementation phases.
- Scope creep, too many changes to the project scope, can hurt the timely completion of the different development phases
- Developers, analysts, and stakeholders may not be available when needed, delaying the project schedule

Political Risks:

- Stakeholders do not buy into or have different goals and/or priorities with the implementation of the COS.
- Management priorities change mid-project, impacting the COS development.

Solution

This section provides a detailed requirements list, the Data Flow Diagram (DFD) of the system, the feasibility analysis including the Net Present Value (NPV) and the return on investment calculations, as well as the breakeven point calculation with a line chart to support your breakeven results, and a the selected development methodology.

Detailed Requirements List

The following requirement list is composed of the requirements provided by HPS management and the requirements gathered by high-level WBS (Work Breakdown Structure) process.

1- Functional Requirements

Functional requirements define what the system needs to do. Functional requirements are often categorized into components representing various processes of the system. For this project, the processes are defined as Customer Account Management, Product Catalog & Browse, Order Placement & Shopping Cart, Checkout Process, Inventory Management Interface, Order Fulfillment Interface, and Administrative Functions.

- Customer Account Management:
 - Allow new customers to create an account online.
 - Allow existing customers to log in to their accounts.
 - Allow customers to view and edit their profile information.
 - Allow customers to view their order history.
 - Provide password recovery methods.
- Product Catalog & Browse:
 - Display product categories and individual products with descriptions, images, and prices.
 - Allow customers to search for products.
 - Allow customers to filter and sort products.
 - Display product information.
- Order Placement & Shopping Cart:
 - Allow customers to add multiple items to an online shopping cart..
 - Allow customers to view and modify items (quantity) in their shopping cart.
 - Calculate and display the total amount for items in the cart.
- Checkout Process:
 - Allow customers to provide or select complete customer information for the order.
 - Calculate order costs based on sales tax, shipping cost, and discounts.
 - Display a final order summary before payment.

- Allow customers to choose from multiple shipping options.
- Process payments via credit card and PayPal.
- Generate an order confirmation number and display order confirmation information.
- Send an order confirmation email to the customer.
- Places a hold (sold tag) on ordered items from the inventory system after the order is paid for.
- Order Fulfillment:
 - Once the order is paid, the order information (customer information, items ordered, shipping information) is sent to the B2C-IS shipping system.

2- Reporting Requirements

These requirements were defined during the documentation requirements gathering process.

- For customers, an order history option needs to be available for viewing, allowing them to view each order individually.
- For HPS staff, sales, customers, and order status reports can be requested and/or viewed.

3- Stakeholder Requirements

Stakeholder requirements are the needs, expectations, and constraints of the different stakeholders gathered during the requirements elicitation process.

- Customers
 - User-friendly, intuitive system.
 - Can be accessed by web browsers.
 - Detailed information about product (pricing), shipping, and order status.
 - Secure transactions and personal information protection.
 - Available to create an account and easy checkout.
- HPS Management
 - Supports B2C model and goals.
 - Can provide data for decision-making.
 - Increased sales and high customer satisfaction.
 - Stays within budget.
- Sales and Purchase Department
 - Pricing and promotions are reflected correctly.
- IT Department & Systems Analyst
 - Maintainable, scalable, and secure system.
 - Easy integration with existing systems.
 - Matches documentation.
- Finance Department
 - Correct tax and total calculations.
 - Secure payment processing.
- Warehouse (Inventory Staff)
 - Interface with the shipping system to process order shipping.
- Legal and Compliance Department
 - Strong data privacy regulations.
 - PCI-DSS compliance.

4- Non-Functional Requirements

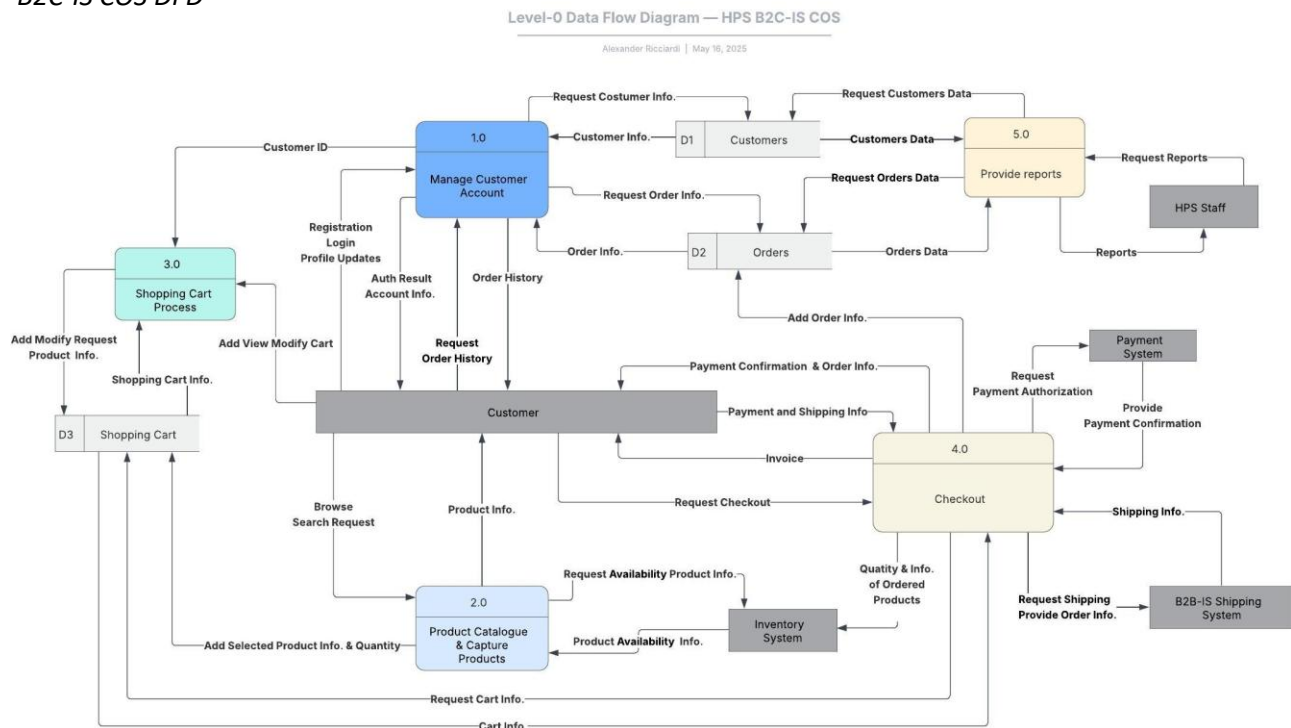
- Performance - fast page loads page loading (3-5 seconds), can handle concurrent user interactions.
- Security – encrypted customer data, processes to protect against web vulnerabilities, and authentication processes need to be implemented.
- Usability – simple navigation and easy to use..

- Reliability/Availability - high uptime (e.g., 99.9%), data backup and recovery.
- Scalability - capable of handling future growth in customers, products, and transactions.
- Maintainability - documented code, easy updates, and maintenance schedule.
- Integrations: Can interface with inventory, finance system, and payment gateways.

Data Flow Diagram

This section provides a high-level Data Flow Diagram (DFD) that was decomposed from the COS context diagram, see Appendix Figure A.1. The high-level DFD illustrates the primary individual processes of the system and their related data flow at the highest level, level-0.

Figure 1
B2C-IS COS DFD



Note: The B2C-IS COS DFD illustrates the primary individual processes of the COS system and their related data flow at the highest level, level-0.

The following table describes the processes illustrated in the high-level DFD. For the data flow, refer to the DFD itself, as they are illustrated by arrows, and for the external entities, refer to the Appendix Table A.1.

Table 1
COS DFD Processes

Process	Description
1.0 Manage Customer Account	This process's role is to manage customers' account information and order history, and order status. It is also responsible for creating, authenticating, and maintaining customer accounts and profiles.

2.0 Product Catalogue & Capture Products	This process's role is to allow customers to "browse / add-to-cart" products by displaying the product's information, such as prices, images, and availability.
3.0 Shopping Cart Process	This process's role is to manage all CRUD operations on the customer's cart. It also displays the shopping cart's contents.
4.0 Checkout	This process's role is to manage checkout by converting the shopping cart's contents into an order after payment is confirmed and the shipping method is selected. It exchanges order information with the Payment System and B2C-IS Shipping System, as well as provides order information to the Inventory System.
5.0 Provide Reports	This process's role is to create and provide customer and order data reports to the HPS staff.

Note: The table describes the different high-level B2C-IS COS processes.

Data Stores

- D1 Customer stores customer account data such as credentials, addresses, and profile.
- D2 Orders stores order data such as products, delivery status, and shipping details.
- D3 Shopping Cart, it is a transient store holding each customer's cart information and state until checkout is performed or a timeout on the store products is reached; purged after order completion.

Feasibility Analysis

A feasibility analysis was conducted to evaluate B2C COS's economic viability. It was performed based on the following provided information:

- Monetary benefits of customer ordering: \$185,000 per year
- One-time development costs: \$275,000.
- Recurring operational costs: \$85,000 per year.
- Discount rate: 12 percent.
- Project lifecycle/time horizon: 5 years.

The calculations were performed based on the cost-benefit analysis techniques: Net Present Value (NPV), Return on Investment (ROI), and Break-Even Analysis (BEA). See Appendix Table A.2 for the technique's descriptions and the Appendix Feasibility Analysis section for more information about the calculations.

Table 2

Feasibility Analysis Spreadsheet

Feasibility Analysis HSP B2C-IS COS						
Cost of Capital	12%					
Initial Capital Expense	\$275,000					
Years		1	2	3	4	5
Cash Inflows (Sales)		\$185,000	\$185,000	\$185,000	\$185,000	\$185,000
Present Value (PV) of Cash Inflows		\$165,179	\$147,481	\$131,679	\$117,571	\$104,974
Cumulative Cash Flow Inflows		\$165,179	\$312,659	\$444,339	\$561,910	\$666,884
Initial Investment	\$275,000					
Cash Outflows (Expenses)		\$85,000	\$85,000	\$85,000	\$85,000	\$85,000
Present Value (PV) of Cash Outflows	\$275,000	\$75,893	\$67,761	\$60,501	\$54,019	\$48,231

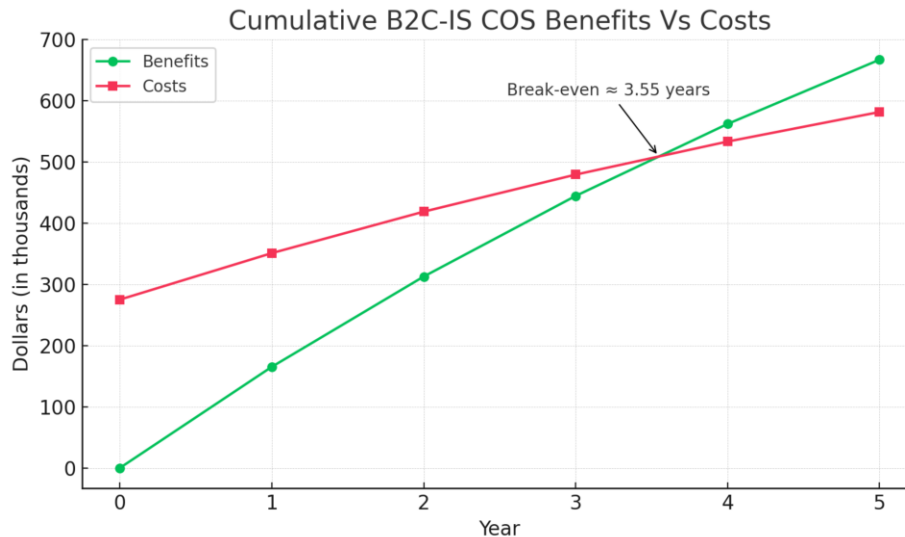
Cumulative PV Cash Outflows (Cumulative Cost)	\$275,000	\$350,893	\$418,654	\$479,156	\$533,175	\$581,406
NPV (Net Present Value)		(\$185,714)	(\$105,995)	(\$34,817)	\$28,735	\$85,478
ROI		-52.93%	-25.32%	-7.27%	5.39%	14.70%
Breakeven Analysis						
Yearly NPV Cash Flow	(\$275,000)	\$89,286	\$79,719	\$71,178	\$63,552	\$56,743
Overall NPV Cash Flow	(\$275,000)	(\$185,714)	(\$105,995)	(\$34,817)	\$28,735	\$85,478

Note: The table illustrates an Excel spreadsheet illustrating the economic feasibility calculations for the five-year project horizon.

As shown in Table 1, Cumulative NPV turns positive part-way through Year 4 and breaks even ≈ 3.55 years, as the interpolating $(|-34\,817| / (63\,552)) = 0.55$ gives the break-even point at about 3.55 years, see the Cumulative B2C-IS COS Discounted Benefits Vs Costs chart, Figure 2. The chart is based on the NPV (Benefits) and the Cumulative PV Cash Outflows (Costs) results from Table 2.

Figure 2

Cumulative B2C-IS COS Discounted Benefits Vs Costs



Note: The figure illustrates the benefits vs costs of the COS over five years based on the feasibility analysis calculation results.

As shown in Figure 2, the two curves intersect at ≈ 3.55 years, confirming the break-even point calculated from Table 2. Other notable results from Table 2 are:

- Total PV Benefits: \$666,884
- Total PV Costs: \$581,406
- Overall NPV: \$ 85,478 (positive $\Rightarrow$ economically attractive)
- Overall ROI: $\text{NPV} \div \text{PV Costs} = 14.7\%$

Together, these calculations results show that the project has excellent economic feasibility, as the project not only pays for itself within $3\frac{1}{2}$ years but also shows a significant benefits margin thereafter.

Development Methodology

To develop the B2C-IS COS is crucial to choose the best-suited development methodology. When considering the overall project as a new B2C e-commerce system, and the COS being just one module of

that system, utilizing the Agile Scrum approach is proposed to develop the system. The Agile methodology is “a project management framework that breaks projects down into several dynamic phases, commonly known as sprints” (Laoyan, 2025, p.1). Scrum is a style of Agile methodology that is suitable for small teams where scrum teams meet daily to discuss active tasks. The Agile Scrum approach is the best suited for the business size of HPS, its iterative nature allows flexibility, uncertainty, and modularity that are essential characteristics for developing a new system and integrating it into a large system, such as the HPS B2C e-commerce system. utilizing the Agile Scrum methodology is proposed to develop the new B2C-IS COS at HPS.

COS Implementation Plan

The COS implementation will follow the proposed Agile Scrum methodology, that is, the development and testing of the project code will need to occur in stages where the COS will be tested and shipped in two-week increments defined as sprints. As HPS operates thirteen nationwide locations and the process implies a complex strategic shift from B2B to a B2C model, the recommended installation strategy is a phased installation. This implies that the core functionalities of the system, such as customer account management and product browsing functionalities, need to be deployed first, followed by ordering and payment processes, and then by more features. Additionally, it is also recommended that the deployment be done using a single-location pilot approach. In other words, the system initially needs to be deployed at one of the HPS branches, followed by a phased rollout to the other 12 sites. This will allow the pilot branch, combined with a selected group of HPS employees and customers, to act as a testing site/group. This approach mitigates risks and captures benefits, as problems can be isolated and resolved before a wider release. The phase rollout to the remaining 12 sites will mitigate the risks of challenges associated with a large-scale deployment, such as service disruptions. Below is a table that details the sprints/phases of the implementation.

Table 3

COS Implementation Sprints/Phases

Sprint / Phase	Target Site(s)	Scope & Major Deliverables	Key Implementation Activities	Primary Roles	Time-box	Entry Criteria	Exit / Sign-off Criteria
Sprint 0 Foundations		- Development, test, staging & production environments - Core DevOps pipeline - Cleansed product-catalog seed data	- Provision servers / cloud resources - Configure CI/CD & security baselines - Extract-transform-load (ETL) initial inventory & pricing	IT Ops, Dev Leads, Security	2 weeks	Budget released; hardware & licenses procured	Environments pass smoke tests; data-migration script validated
Sprint 1 Account & Catalog	Pilot branch (internal only)	- Customer account module (create / log-in / profile) - Product browse, search, filters	- Code & unit-test features- - Integration test with inventory API (read-only)- - UAT “happy path” for account creation & browsing	Dev Team, QA, Marketing SMEs	2 weeks	Sprint 0 artifacts deployed to DEV	UAT sign-off (Marketing & Sales); backlog defects < Level 3

Sprint 2 Cart, Tax & Checkout	Pilot branch	- Shopping-cart services - Sales-tax calc engine - Order summary UI	- Integrate cart with account & catalog- Mock payment gateway for sandbox testing - Regression of Sprint 1	Dev Team, QA, Finance	2 weeks	Sprint 1 accepted in STAGE	Secure code review passed; Finance approves tax logic
Sprint 3 Payment & Inventory Debit	Pilot branch	- Real payment-gateway connector (credit & PayPal) - Real-time stock decrement	- PCI-scope pen-test- End-to-end order flow in STAGE- Alpha UAT (internal staff)	Dev, QA, Security, Finance, Warehouse	2 weeks	Payment sandbox credentials; inventory stub ready	Alpha UAT exit criteria met; critical bugs resolved
Sprint 4 Shipping Interface & Reporting	Pilot branch	- API push to B2C-IS Shipping - Basic operational reports	- Message-queue or REST hookup- Verify label generation & status callbacks - Build daily sales / stock report	Dev, QA, Warehouse, IT Ops	2 weeks	Sprint 3 accepted; Shipping API available	End-to-end "order-to-dispatch" succeeds; Ops sign-off
Pilot Go-Live (Hyper-sprint)	1 HPS branch + select online customers	- Full COS in production for pilot cohort	- Quiet launch during off-peak- 24 x 7 hyper-care & telemetry - Collect user feedback & KPIs	IT Ops, Dev on-call, Pilot CSRs	4 weeks	Sprints 0-4 completed; Go/No-Go board approves	≥ 99.5 % uptime; KPI thresholds met; lessons-learned doc
Wave 1 Roll-out	Next 3-4 branches	- Same feature set + refinements	- Incremental data migration - Staff training sessions - Update SOPs based on pilot lessons	IT Ops, Training Leads, Local Mgmt	2 weeks	Pilot retros resolved; training content ready	Branch managers sign-off; no Severity-1 issues after 72 h
Wave 2 Roll-out	Remaining 9 branches	- Enterprise-wide availability	- Cut-over remaining sites - Decommission legacy B2B workarounds for consumer orders	IT Ops, Change-Mgmt, Exec Sponsor	2 weeks	Wave 1 stable; capacity tests passed	COS live at all 13 sites; legacy workflows retired
Post Go-Live & Handover	All sites	- Knowledge-transfer to Support - Final documentation & project closure	- Hyper-care ↓ to normal SLA - Archive code & infra artefacts - Close vendor contracts & budget	Project Mgr, Support Lead, Finance	2 weeks	Wave 2 accepted; support team staffed	Support tickets within BAU range; project closure sign-off

Note: the table lists the different sprints and phases of the COS implementation. Sprints are two weeks long, but they can be iterated several times until their goals are achieved or an exit criterion is met. This is also true for the wave roll-outs.

COS Testing and Training Plans

In conjunction with the implementation plan, the testing and training plans described below ensure that the COS meets the project requirements, it is a robust system, and that HPS staff and customers can understand and use the system effectively.

Testing Plan

The testing plan follows a testing strategy that ensures that the system functions as intended, meets the project's requirements, and is free of bugs, easy to use, and secure. To meet these goals, the plan implements several test levels and types such as unit testing, integration testing, system testing, and user acceptance testing (UAT). See Table 4 for a detailed description of the testing plan.

Table 4
COS Test Plan

Test Type	Objectives	Methods/Techniques	Tools (Examples)	Responsible Department/Team
Unit Testing	Test individual COS modules (tax calculation, cart item management) - need to function correctly in isolation.	Code reviews, developer-written tests, automated test scripts.	JUnit (Java), NUnit (.NET), PyTest (Python)	Developers
Integration Testing with the Payment System	Test COS transaction processing with Credit Card and PayPal gateways. Test if the correct data exchange for authorization and confirmation.	API testing, end-to-end transaction scenarios, error handling tests.	Postman, SoapUI, custom scripts	Developers, Testers
Integration Testing with the Inventory System	Test COS product management - real-time product availability, updates, and removal from inventory stock	API testing, data synchronization checks, scenario-based tests (e.g., low stock).	Custom scripts, database query tools	Developers, Testers
Integration Testing with the Shipping System	Test the COS's timely transfer of order details (customer info, items, shipping address) to the B2C-IS Shipping System.	API testing, message queue validation (if applicable), data integrity checks.	TBD (depends on shipping system interface)	Developers, Testers
Integration Testing with the Entire B2C System	Test COS with the end-to-end B2C business processes (account creation, multi-item order, checkout, confirmation) meet all functional requirements.	Scenario-based testing, user story validation, exploratory testing.	Manual testing, Selenium (for UI automation)	Testers, Business Analysts
System Testing - Security	Test COS protection of customer data (encryption), secure authentication, and resilience against common web vulnerabilities.	Penetration testing (basic), vulnerability scanning, security code reviews.	OWASP ZAP, Burp Suite (Community Edition)	Testers, Security Specialist (if avail.)
System Testing - Performance	Test if COS meets page load time targets (3-5 seconds), handles expected concurrent user load, and maintains responsiveness under stress.	Load testing, stress testing, endurance testing, page speed analysis tools.	Apache JMeter, LoadRunner, Google PageSpeed	Testers, Performance Engineers

UAT - Alpha Testing	Internal HPS staff (Marketing, Sales, IT, Finance, Warehouse) validate COS meets business needs, is usable, and robust. Includes recovery, stress, and specific business scenario testing.	Scenario-based execution by internal users, feedback sessions, bug reporting.	Test environment, bug tracking tool (e.g., GitHub)	HPS Staff (Marketing, Sales, IT, Finance, Warehouse)
UAT - Beta Testing	Selected external customers/pilot users validate COS in a real-world setting, focusing on usability (user-friendly, intuitive system), customer experience, and functionality.	Real-world usage by pilot users, surveys, feedback forms, direct observation (if pos		

Note: The table showcases the COS test plan by describing the different test types used by the plan, where these tests need to be used (objectives), and who is responsible for implementing them.

Training Plan

The training plan goal is to develop a training program that will be effective for equipping HPS staff with the necessary knowledge and skills to use and maintain the COS, and for guiding customers in its use. A training program is based on the system documentation developed during the multiple phases of the COS development project, including requirements, design, and implementation. Such a program will ensure that HPS staff understand and will prepare to use the new COS properly, especially for HPS customer service staff who are in charge of handling customer questions, resolving order issues, assisting with account management, and providing general support for COS users. Moreover, an effective training program is critical to user adoption. Studies have shown that 70% of organizations face delays or failures in system implementations because of poor training and user adoption (Gendron, 2024). Therefore, a training plan is a crucial factor for the successful implementation of the new COS. The COS training plan is composed of various plan modules targeting different COS users or stakeholders, such as COS Customer Account & Order Management training and Financial Processing & Reporting in COS training, respectively targeting Customer Service Representatives - Sales Staff and Finance Dept. - Accounting Staff. The table below provides a detailed description of the different training modules that are part of the training plan.

Table 4
COS Training Modules

Training Module	Target HPS Staff Group(s)	Objectives
Introduction to COS & HPS B2C Model	All relevant HPS Staff	Understand B2C strategy, COS purpose, key features, impact on HPS operations, and changes to individual roles.
COS Customer Account & Order Management	CSRs, Sales Staff	Create/manage customer accounts, process orders, handle inquiries, manage returns/exchanges, understand order status tracking.
Product Catalog & Promotion Management	Marketing Dept, Sales Dept (select staff)	Add/edit products & categories, manage pricing & images, create/manage promotions and discount codes within COS.
Inventory Interface & Fulfillment Coordination	Warehouse Staff, Inventory Control	Understand how COS orders impact inventory, process pick lists generated by COS, update order status for shipping, coordinate with shipping system.

Financial Processing & Reporting in COS	Finance Dept, Accounting Staff	Process payments/refunds, reconcile sales data, understand tax calculations & reporting, generate financial reports from COS.
COS System Administration & IT Support	IT Department	Perform system monitoring, user access control, basic troubleshooting, data backup/recovery procedures, understand system architecture.
Advanced Reporting & Analytics	Management, Marketing Analysts, Sales Managers	Utilize advanced reporting features, interpret customer behavior analytics, generate custom reports for decision-making.

Note: The table provided a detailed list of the different COS training modules, including their target HPS staff or group and their descriptions/objectives.

Recommendations/Conclusions

Hullabaloo Party Supplies (HPS) has decided to transition from its traditional B2B sales model to a direct B2C model. This proposal examines the COS subsystem of the overall B2C-IS that HPS wants to implement. The COS core functionalities allow individual customers to browse products, create accounts, place orders, make payments, and choose shipping options. The proposal states the goal and the objective of HPS's new COS, which is to transition from their existing B2B-IS to a new B2C-IS as the existing system does not meet the requirements of the B2C business model and the new COS needs to function as a core B2C-IS component meeting the requirements provided by HPS management. The proposal identifies that the main issue with the existing system is that B2B-IS is optimized for large-volume transactions with vendors and lacks the architecture, user interface, and functionalities required for a B2C e-commerce small-volume consumer transactions. The proposal has also provided a list of assumptions covering resource availability, stakeholder commitment, scope stability, and technical integrations. Additionally, the proposal identifies the main stakeholders and provides a high-level WBS for requirements Gathering. It also identifies the main risks associated with the development and implementation of COS and provides a comprehensive list of those risks.

The proposal also provides a solution to COS by detailing a list of the system requirements, that is, the functional and non-functional requirements. By illustrating the system architecture using a Data Flow Diagram (DFD) and a table describing the DFD processes. The proposal's feasibility analysis of the COS shows that the project aligns strongly with HPS's strategic objective, which is supported by an NPV \$85,478, ROI 14.7%, breakeven ~3.55 years, illustrating the financial viability of the COS project. Additionally, the proposal recommends using the Agile Scrum development methodology for developing, implementing, and maintaining the system. To support further, the COS solution, the proposal provides COS implementation, testing, and training plans, which are crucial for the successful implementation and deployment of the system.

In conclusion, the COS aligns well with HPS's shift from a B2B to a B2C model. As shown in this proposal, the project has some risks. However, by using the system architecture illustrated by the Data Flow Diagram, by adopting the Agile Scrum development methodology for developing, implementing, and maintaining the system, and by following the provided implementation, testing, and training plans, the project is predicted to be very successful. This is supported by feasibility analysis showing positive NPV and ROI projections (NPV = \$85,478; ROI = 14.7%) and a breakeven at approximately 3.55. Therefore, HPS can develop and implement a COS project with confidence as it will deliver significant value and will allow HPS to be successful in its new endeavor in the B2C e-commerce field.

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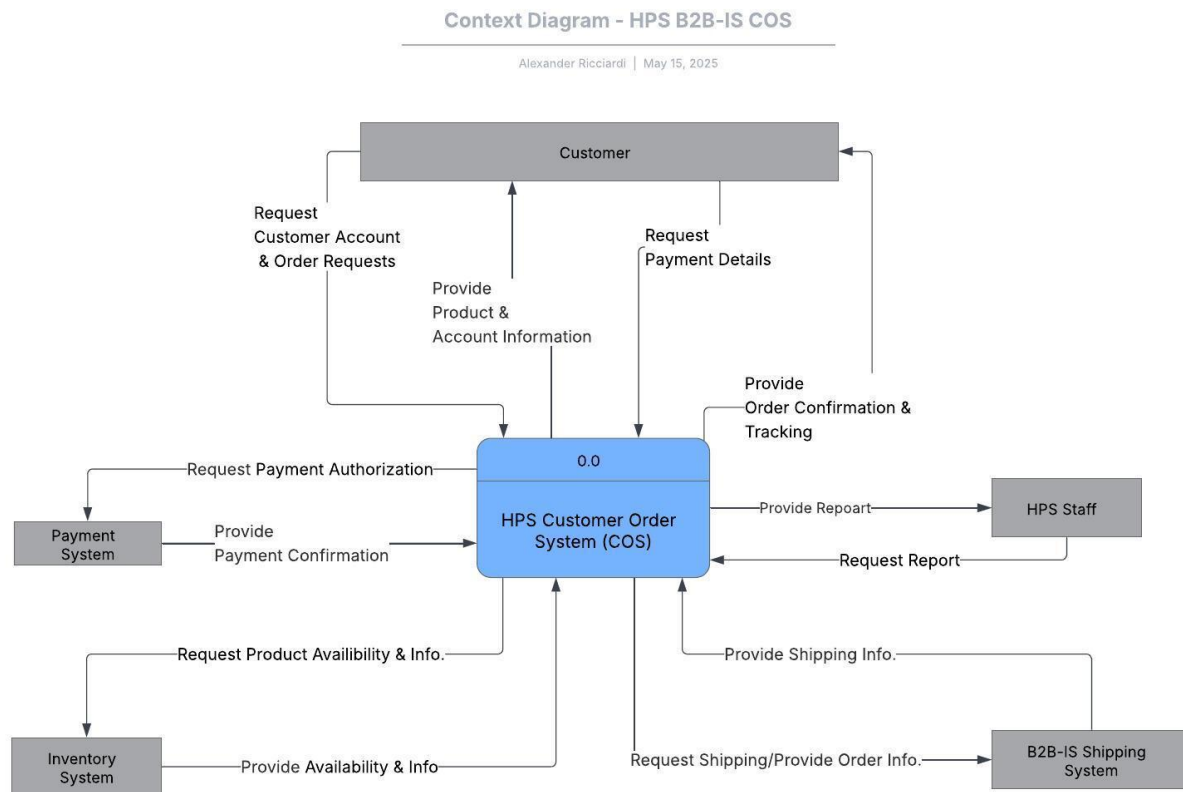
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Appendix

Miscellaneous Figures and Tables

Figure A.1
COS Context Diagram



Note: The context diagram illustrates the highest view of the HPS B2C-IS COS data flow.

Table A.1
COS DFD External Entities

External Entity	Role in the System
Customer	It is an actor/external stakeholder; it is the end-user.
Payment System	This system is a third-party payment-gateway service that authorizes credit-card and PayPal payments.
Inventory System	This system is an internal HPS IS responsible for managing the product inventory.
B2C-IS Shipping System	This system is an internal HPS IS internal system responsible for product shipping.
HPS Staff	It is an actor; it is an internal stakeholder (sales, support, finance).

Note: The table describes external B2C-IS COS entities that interact with the system.

Feasibility Analysis

Table A.2

Cost-Benefit Analysis Techniques

Analysis Technique	Description
Net Present Value (NPV)	NPV uses a discount rate determined from the company's cost of capital to establish the present value of a project. The discount rate is used to determine the present value of both cash receipts and outlays.
Return on Investment (ROI)	ROI is the ratio of the net cash receipts of the project divided by the cash outlays of the project. Trade-off analysis can be made among projects competing for investment by comparing their representative ROI ratios.
Break-Even Analysis (BEA)	BEA finds the amount of time required for the cumulative cash flow from a project to equal its initial and ongoing investment.

Note: The table describes commonly used cost-benefit analysis techniques. Figure 5.7, from “Chapter 5: Initiating and Planning Systems Development Projects. Modern systems analysis and design (9th ed.)” by Valacich and George (2020a, p.121)

Table A.3

Calculation Formulas

Worksheet Row / Concept	Formula	Description
Present Value (PV) of a single cash flow	$PV_n = \frac{Y_n}{(1+r)^n}$	Y_n = cash amount in year n r = discount rate
Cumulative PV of Inflows	$CumPV_n = CumPV_{n-1} + PV_n$	Total Inflows Cash
Year-0 PV Initial Investment	$PV_0 = I_n$	I_0 = initial cash outflow
PV of Recurring Expense	$PVCost_n = \frac{C_n}{(1+r)^n}$	C_n = expense in year n
Cumulative PV of Costs	$CumCost_n = CumCost_{n-1} + PVCost_n$	Total Cost
Net Present Value (NPV)	$NPV = \sum_{n=0}^N (PV_n - PVCost_n) \Rightarrow$ $NPV = CumPV_N - CumPVCost_N$	Total benefit minus total cost
Return on Investment (ROI)	$ROI = \frac{NPV}{CumPVCost_N}$	Ratio of net value to total cost
Yearly NPV Cash Flow	$\Delta NPV_n = PV_n - PVCost_n$	Net cash in year n
Overall (Cumulative) NPV Cash Flow	$CumNPV_n = CumPV_n + \Delta NPV_{n-1}$	Break-even year is where $CumNPV_n$ first ≥ 0

Note: The table provides a list of the various formulas used to conduct the feasibility economic analysis.

For the following table, refer to the Excel spreadsheet file, *Feasibility Analysis HPS.xlsx*

Table A.4

Feasibility Analysis Excel Formulas

Cell	Formula	Description
C6	=C5/(1+\$C\$2^C\$4	PV of Year 1 inflow
D6	=D5/(1+\$C\$2^D\$4	PV of Year 2 inflow
E6	=E5/(1+\$C\$2^E\$4	PV of Year 3 inflow
F6	=F5/(1+\$C\$2^F\$4	PV of Year 4 inflow
G6	=G5/(1+\$C\$2^G\$4	PV of Year 5 inflow
C7	=C6	Cumulative PV inflows Yr 1
D7	=C6+D7	Cumulative PV inflows Yr 2
E7	=D7+E6	Cumulative PV inflows Yr 3
F7	=E7+F6	Cumulative PV inflows Yr 4
G7	=F7+G6	Cumulative PV inflows Yr 5
B10	=B8	PV cost in Year 0
C10	=C9/(1+\$C\$2^C\$4	PV of Year 1 expense
D10	=D9/(1+\$C\$2^D\$4	PV of Year 2 expense
E10	=E9/(1+\$C\$2^E\$4	PV of Year 3 expense
F10	=F9/(1+\$C\$2^F\$4	PV of Year 4 expense
G10	=G9/(1+\$C\$2^G\$4	PV of Year 5 expense
B11	=B10	Cumulative PV costs Yr 0
C11	=B11+C10	Cumulative PV costs Yr 1
D11	=C11+D10	Cumulative PV costs Yr 2
E11	=D11+E10	Cumulative PV costs Yr 3
F11	=E11+F10	Cumulative PV costs Yr 4
G11	=F11+G10	Cumulative PV costs Yr 5
G12	=G7-G11	NPV (equivalent equation for cells C12 to F12)
G13	=G12/ABS(G11)	ROI (equivalent equation for cells C13 to F13)
B15	=-B10	Yr 0 yearly NPV cash flow
C15	=C6-C10	Yr 1 yearly NPV cash flow
D15	=D6-D10	Yr 2 yearly NPV cash flow
E15	=E6-E10	Yr 3 yearly NPV cash flow
F15	=F6-F10	Yr 4 yearly NPV cash flow

G15	=G6-G10	Yr 5 yearly NPV cash flow
B16	=B15	Cumulative NPV Yr 0
C16	=B16+C15	Cumulative NPV Yr 1
D16	=C16+D15	Cumulative NPV Yr 2
E16	=D16+E15	Cumulative NPV Yr 3
F16	=E16+F15	Cumulative NPV Yr 4
G16	=F16+G15	Cumulative NPV a Yr 5

Note: The table provides a list of the various formulas used to conduct the feasibility economic analysis.